

Advika Namasivayam

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EDUCATION

Massachusetts Institute of Technology (MIT) | Cambridge, MA

Expected May 2028

Bachelor of Science in Mechanical Engineering; Minor in Computer Science and Engineering

GPA: 4.9/5.0

Relevant Coursework: Dynamics & Control I-II; Mechanics & Materials I-II; Design & Manufacturing I-II; Thermal-Fluids Engineering I-II; Space Systems Engineering; Mobile Autonomous Systems Laboratory (MASLAB); Numerical Computation; Linear Algebra & Optimization

SKILLS

- **Design & Analysis:** PTC Creo, Siemens NX, SolidWorks, ANSYS (Mechanical & Discovery), Fusion 360, Onshape, Rhino
- **Manufacturing:** CNC/Manual machining, TIG welding, waterjet & laser cutting, FDM, SLA, EOS metal additive manufacturing
- **Programming & Controls:** Python, MATLAB, Simulink, C, Arduino, LabVIEW, Java, Git, GitHub

EXPERIENCE

Engineering Intern - Additive Manufacturing | Ingersoll Rand | Quincy, IL

05/26–08/26

- Designed mechanical components and tooling using PTC Creo and Siemens NX for additive manufacturing applications
- Supported EOS metal additive manufacturing workflows from DfAM through build preparation and dimensional validation
- Applied ANSYS Discovery to evaluate and optimize heat exchanger designs for thermal performance and manufacturability
- Designed and executed a DOE to optimize SLA printing parameters, improving dimensional accuracy and print quality

MIT Solar Electric Vehicle Team | Cambridge, MA

Captain

05/26 - Present

- Lead a 70-member interdisciplinary engineering team developing next-generation solar vehicles targeting the 2028 and 2030 American Solar Challenge
- Direct technical strategy, subsystem integration, design reviews, fabrication planning, and vehicle development across mechanical, electrical, and aerodynamics teams

Vice-Captain & Roll Cage Lead

09/24 - 05/26

- Led the design and fabrication of a TIG-welded steel occupant safety structure for *Solstice*, the team's 2026 Formula Sun Grand Prix vehicle.
- Performed CAD and ANSYS FEA to validate structural integrity and optimize roll cage geometry, reducing mass by 25% compared with that of the team's previous vehicle
- Coordinated fabrication and cross-subsystem integration to meet competition safety and structural requirements

Machine Learning for CAD Researcher | MIT DeCoDE Lab | Cambridge, MA

06/25–08/25

- Developed Python and Rhino workflows to generate B-spline surfaces and repair CAD meshes, improving geometric accuracy while implementing genetic algorithms for physically consistent ship-hull optimization

Robotics Researcher | MIT Tasean Group | Cambridge, MA

01/25–05/25

- Designed and implemented a laser and robotic arm control system in LabVIEW, integrating precise motion-planning techniques to improve precision in metal microstructure manipulation

Makerspace Mentor | MIT | Cambridge, MA

06/25–Present

- Trained students on machining, fabrication, and safe operation of makerspace equipment

PROJECTS

Vine Robot Closed-Loop Control | MIT d'Arbeloff Lab

06/25–01/26

- Developed a closed-loop control system for a pressure-driven vine robot, optimizing actuator behavior to improve curvature control and navigation precision

MakeMIT Hardware Hackathon: Home Security Track Winner | MIT

02/26

- Designed and fabricated “knobbi,” a retractable doorknob with a custom tri-bar linkage and autolock features
- Integrated mechanical hardware with software for user authentication via camera-based facial recognition
- Engineered a flush-mounted retractable housing to reduce pry points and improve forced-entry resistance

Mechanical Subteam Member | MIT Lunabotics

06/25–Present

- Designed mechanical hardware for a NASA Lunabotics Challenge rover, emphasizing DFM and regolith collection

MIT MASLAB: Wilkens Family Award for Most Creative Design | MIT MASLAB 2025

01/25

- Designed and fabricated robot frame, vertical lift, and intake mechanisms for reliable autonomous stacking, sorting, and dumping of game pieces
- Integrated mechanical hardware with ROS and YOLO-based vision, engineering a custom hook-linkage intake and elevator architecture that improved reliability and cycle speed